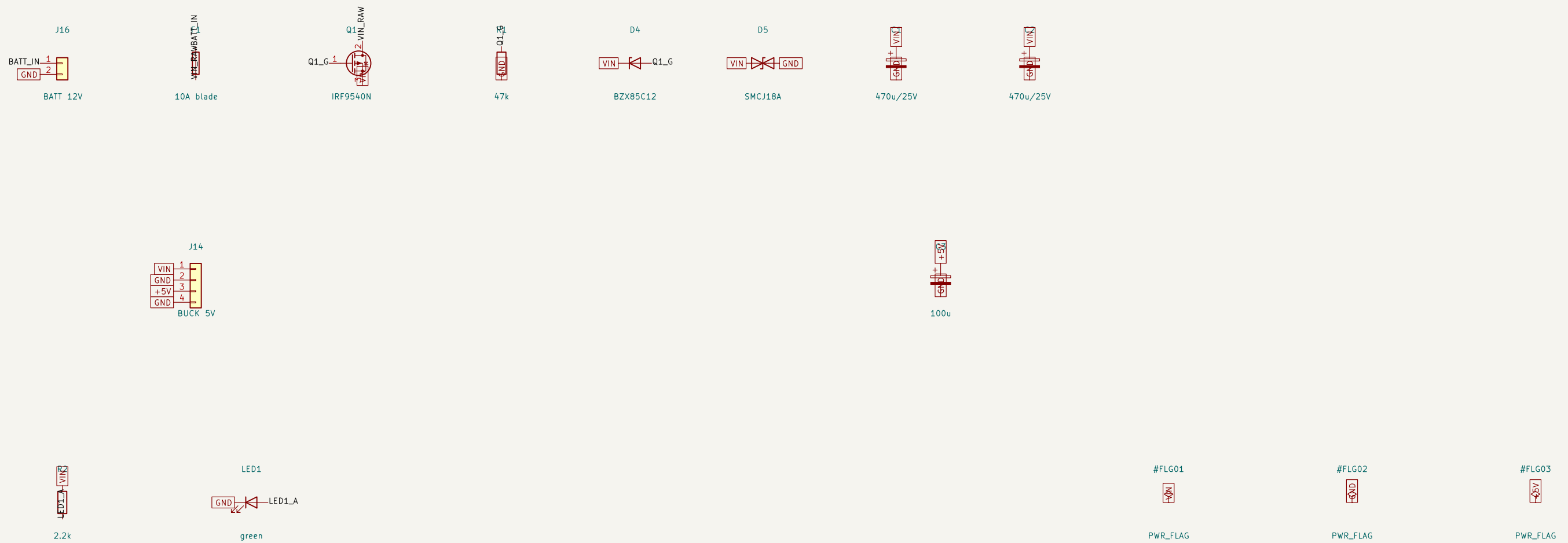




POWER: 12V battery in, protection, generic 5V buck on J14

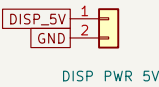
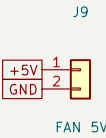
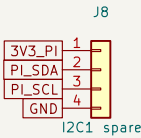
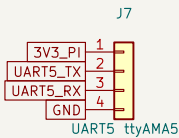
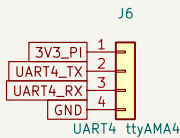
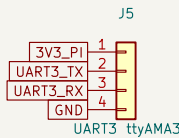
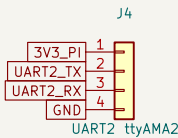
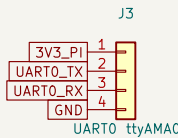
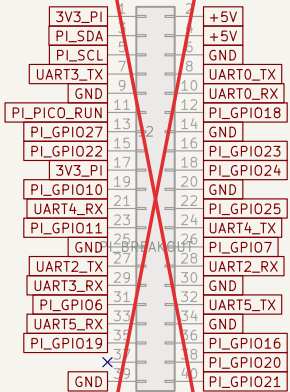
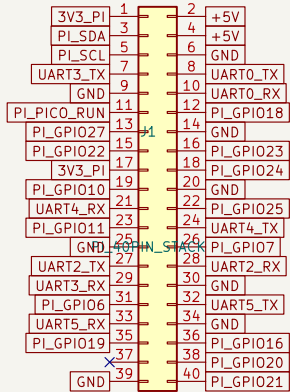
J14: VIN GND 5V GND – wire to an XL4015/LM2596–class module, set 5.1V BEFORE fitting

Servo bridge runs on protected VIN directly (12V-only board)

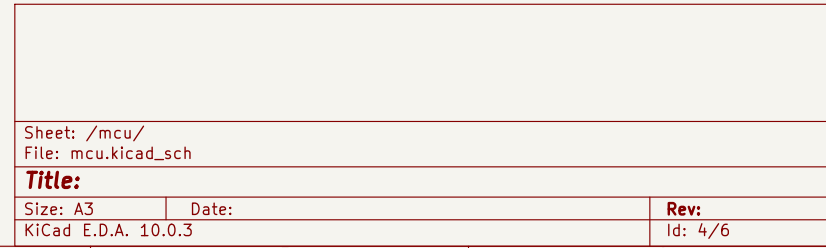


RASPBERRY PI 4/5 CARRIER: stacking header, UART breakouts, display power

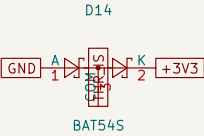
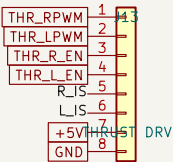
DSI touchscreen: ribbon to Pi directly; 5V from J10. USB/HDMI on Pi itself.



Pico soldered directly (low profile, sits under the Pi). Pi reflashes it via SWD on GPIO24/25 (openocd).



THRUST: control header for external driver (IBT-2 / BTS7960-class)
J13 pinout matches the common IBT-2 module: RPWM LPWM R_EN L_EN R_IS L_IS VCC GND
Driver held disabled by 100k pull-downs unless the Pico actively enables it
Battery/motor power stays on the external driver – only logic on this board



AS5600 (in servo housing per vanchor-ng cad/) cabled to J11; keep lead <1m, 100kHz

24-48V boats: servo stall limited by BUCK2 (4.5A). 12V boats: J17 links VIN->12V, full stall current

